

1) Determinazione specifiche di progetto

Incognite: G_F ; K_c da ε_r ; K_c da d_a e/o d_p ; ω_c da d_p e/o d_s ; ζ , $T_{p,0}$, $S_{p,0}$ da $\hat{s}$; ω_c da t_r ; ω_c da $t_{s,\varepsilon\%}$; ν da d_a e/o d_p ; M_S^{LF} , M_T^{HF} .

a) Calcolo G_F

$$G_F = \frac{1}{G_s K_d}$$

K_p = guadagno di G_p in cost. tempo. (p : poli nell'origine di G_p spostati in cost. tempo)

b) Calcolo ν e K_c da ε_r

se $\varepsilon_r = 0 \rightarrow \nu > h - p$, $|K_c| = \forall$

se $\varepsilon_r \neq 0 \rightarrow \nu \geq h - p$

$$|K_c| \geq \begin{cases} \left| \frac{R_0 K_d^2}{\varepsilon_r K_p G_a} \right| & \text{se } \nu + p \neq 0 \\ \frac{R_0 K_d^2}{\varepsilon_r} - K_d & \text{se } \nu + p = 0 \end{cases}$$

c) Calcolo K_c da d_a

$$|K_c| \geq \frac{|D_{a,0}|}{\varepsilon_{d_a} |G_a G_s G_F|} \quad \text{se } \varepsilon_{d_a} \neq 0 \Rightarrow \nu \geq h_a$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_a} = 0 \Rightarrow \nu > h_a$$

d) Calcolo K_c da d_p (polinomiale) o ω_c/M_S^{LF} (sinusoidale)

Polinomiale:

$$|K_c| \geq \frac{|D_{p,0}|}{\varepsilon_{d_p} |G_a G_s G_F K_p|} \quad \text{se } \varepsilon_{d_p} \neq 0 \Rightarrow \nu \geq h_p - p$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_p} = 0 \Rightarrow \nu > h_p - p$$

Sinusoidale:

$$M_S^{LF} = \frac{\rho_p}{a_p}, \quad M_{S,dB}^{LF} = 20 \log_{10} M_S^{LF}$$

$$\omega_L = \omega_p \cdot 10^{-M_{S,dB}^{LF}/40}, \quad \omega_c \geq 2\omega_L$$

e) Calcolo ω_c/M_T^{HF} da d_s (sinusoidale)

$$M_T^{HF} = \frac{\rho_s G_s}{a_s}, \quad M_{T,dB}^{HF} = 20 \log_{10} M_T^{HF}$$

$$\omega_H = \omega_s \cdot 10^{M_{T,dB}^{HF}/40}, \quad \omega_c \leq \frac{\omega_H}{2}$$

f) Calcolo ζ , $T_{p,0}$, $S_{p,0}$ da $\hat{s}$

$$\zeta \geq \frac{|\ln(\hat{s})|}{\sqrt{\pi^2 + \ln^2(\hat{s})}}$$

$$T_{p,0} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$S_{p,0} = \frac{2\zeta\sqrt{2+4\zeta^2+2\sqrt{1+8\zeta^2}}}{\sqrt{1+8\zeta^2+4\zeta^2-1}}$$

g) Calcolo ω_c da t_r e $\varepsilon_{s,\%}$

$$t_r : \quad \omega_c \geq \left(\frac{\pi - \arccos(\zeta)}{t_r \sqrt{1-\zeta^2}} \right) \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

$$t_{s,\alpha\%} : \quad \omega_c \geq \left(\frac{|\ln(\alpha\%|)}{\zeta t_{s,\alpha\%}} \right) \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

2) Progetto del Sistema di Controllo

Passo 1: Studio del segno di K_c e stabilizzabilità

a. Scegliere un segno e calcolare $L_{in}(s)$:

$$L_{in}(s) = \frac{K_c}{s^\nu} \cdot G_F G_a G_p G_s$$

b. Tracciare il diagramma di Nyquist di $L_{in}(s)$ e calcolare

$P_{cl} = N + P_{ol}$, dove:

N : intersezioni semiretta

P_{ol} : poli con $Re(p_i) > 0$ (di solito nullo)

• P_{cl} pari $\rightarrow$ **Stabilizzabile**

• P_{cl} dispari $\rightarrow$ **Cambio segno di K_c**

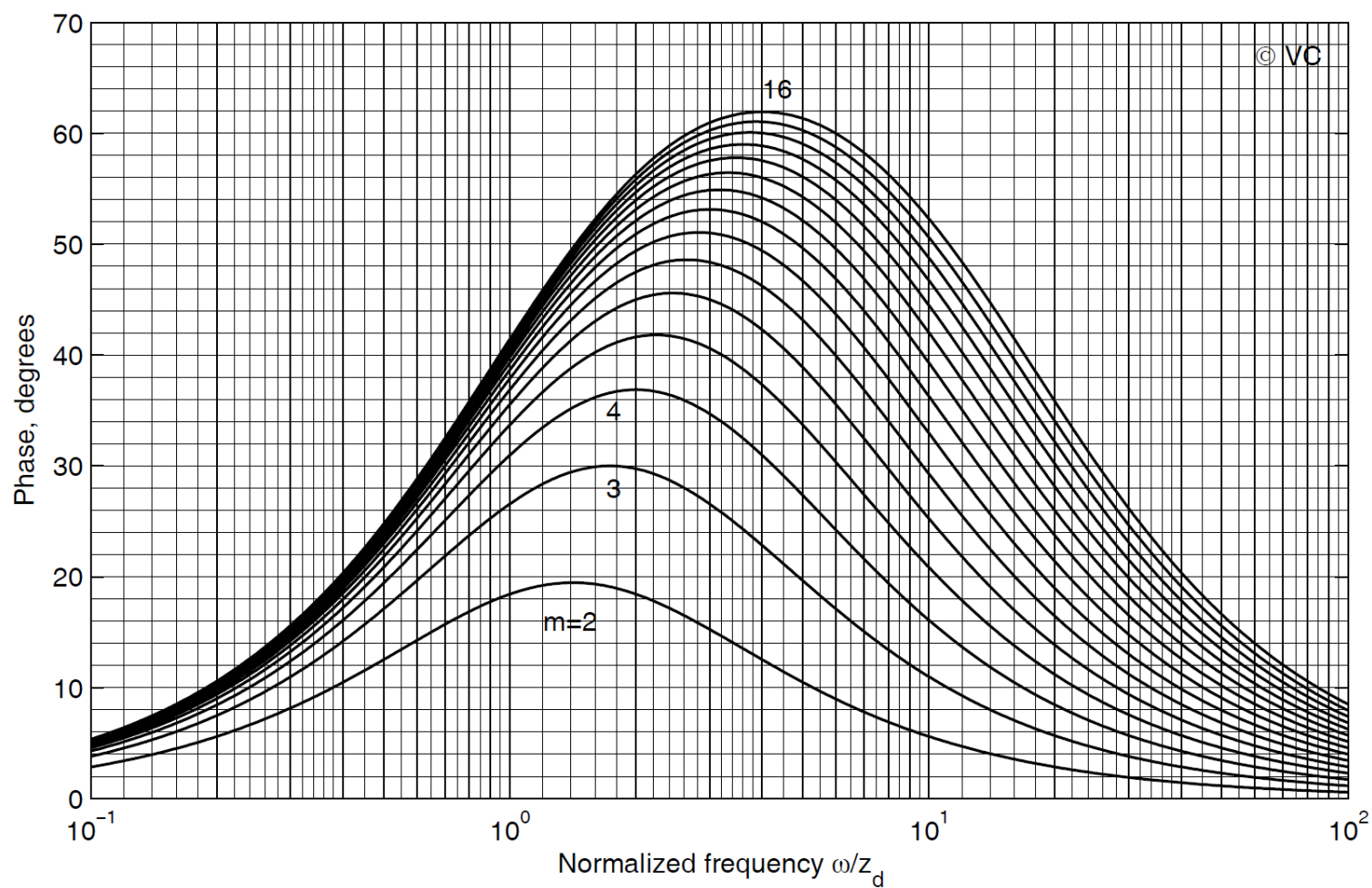
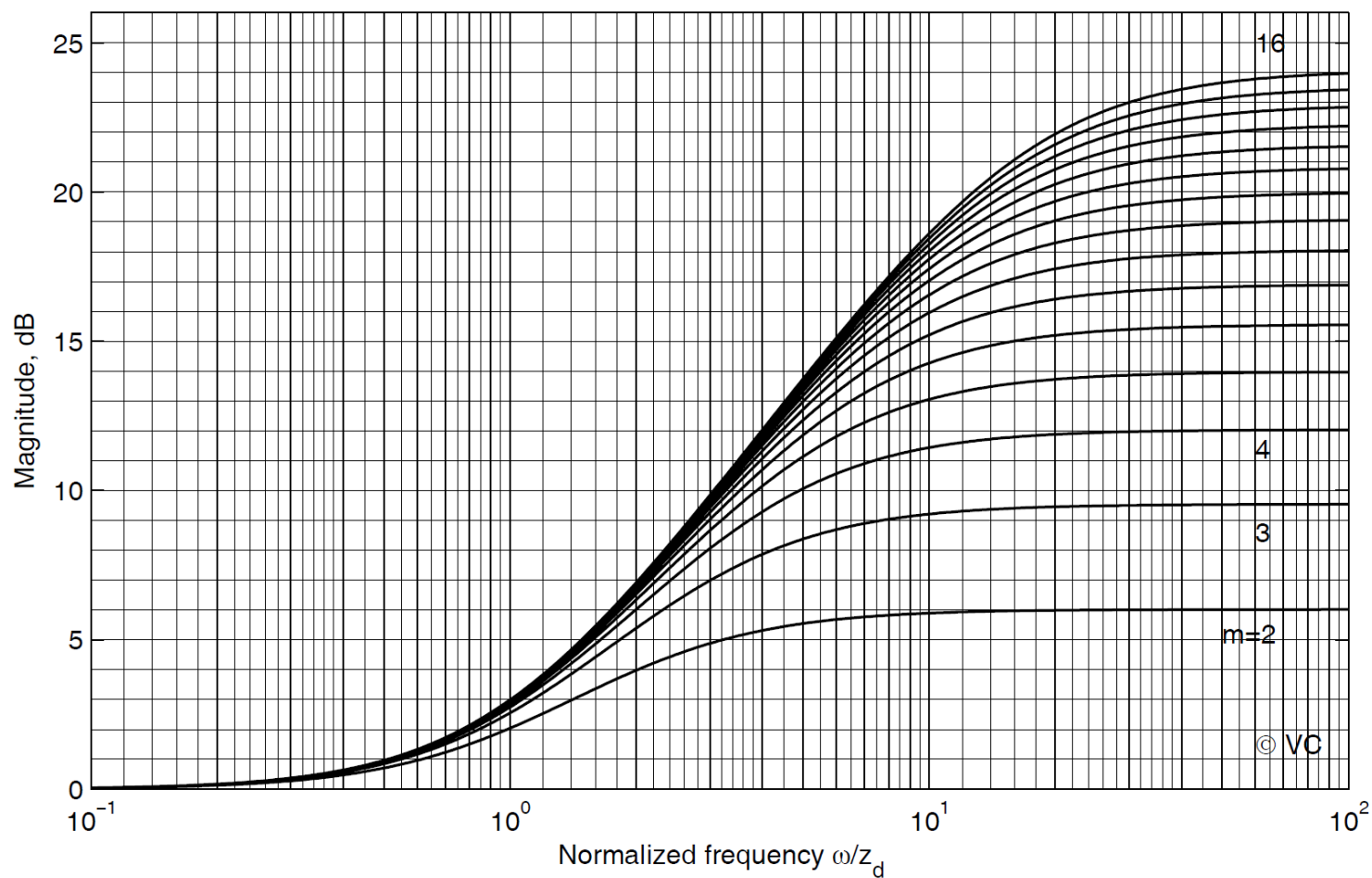
c. Scegliere K_c poco più grande del vincolo (se c'è).

$$[N_{L_{in}}, D_{L_{in}}] = \text{tfdata}(L_{in}, 'v')$$

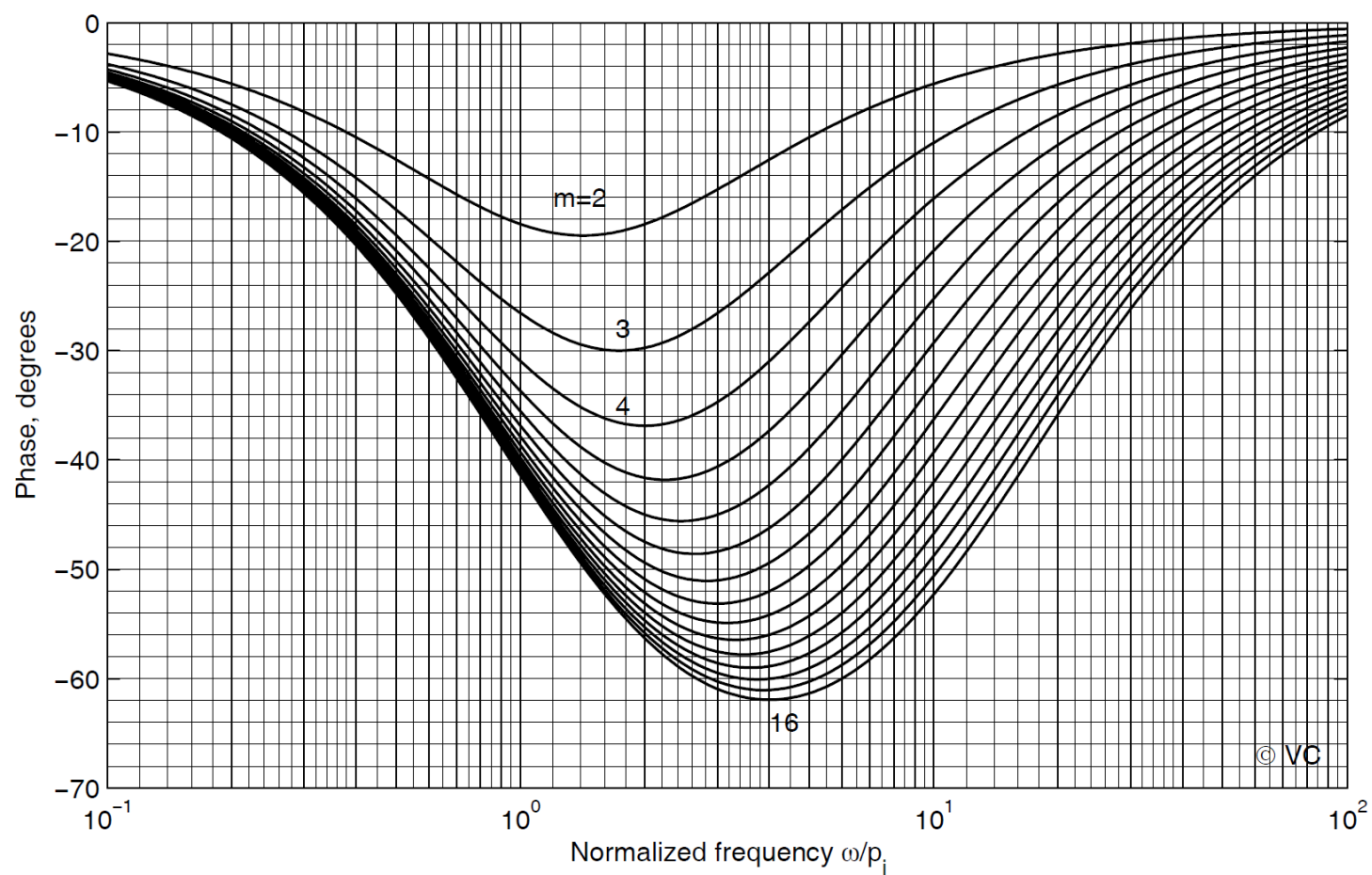
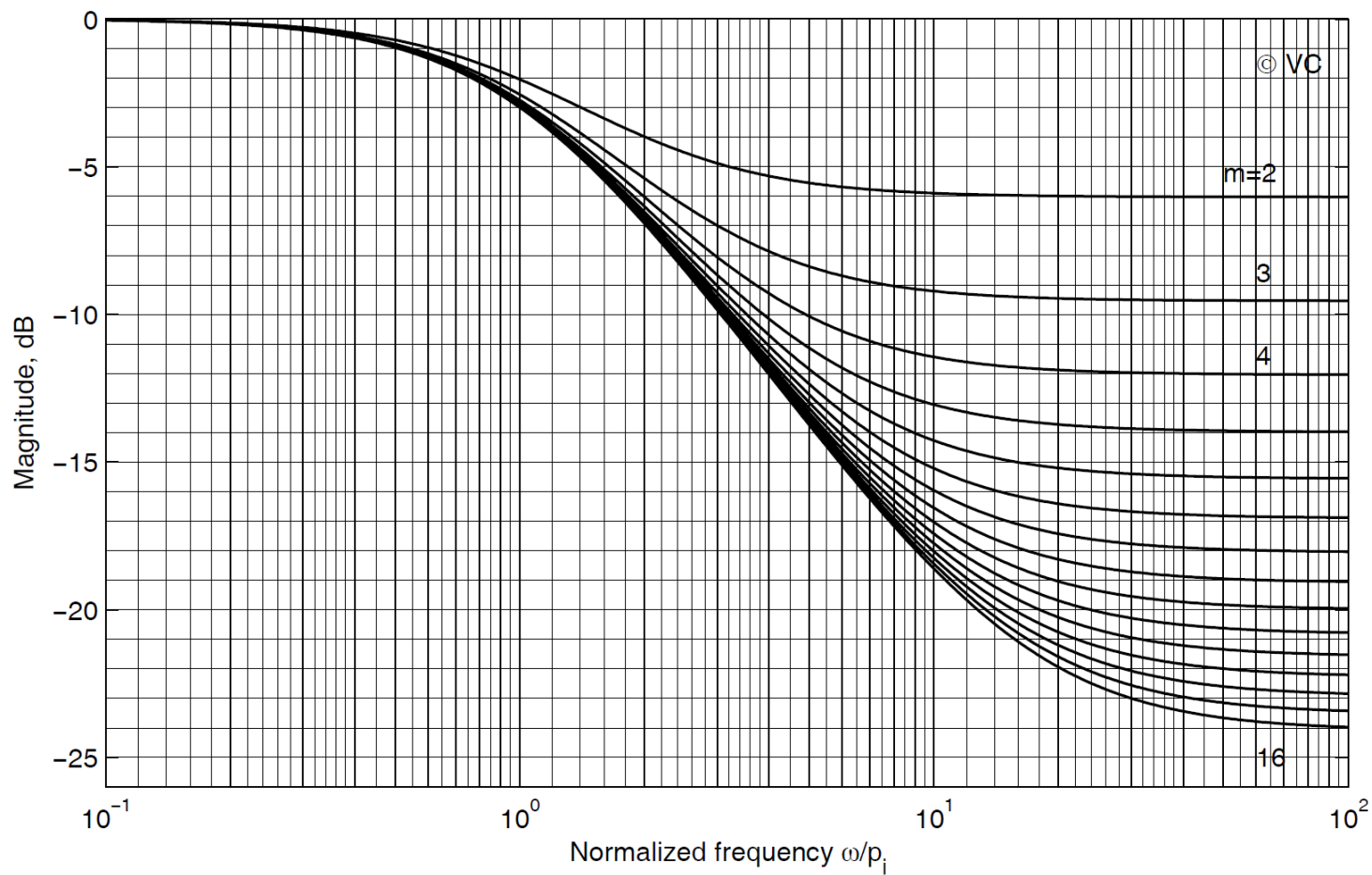
$$\text{mynggridst}(T_{p,0}, S_{p,0})$$

$$\text{nyquist1}(N_{L_{in}}, D_{L_{in}})$$

Phase LEAD network



Phase LAG network



Zero network

